

NorthPay

Canadian Payroll Calculation Reference

Tax year 2026 · 9 provinces · 4 pay frequencies

Every formula, every constant, every bracket used by the NorthPay engine — alongside worked examples so you can verify any paystub or T4 by hand.

Tax year	2026
Provinces supported	ON · AB · BC · MB · SK · NS · NB · PE · NL
Quebec	Intentionally excluded (separate RQ system)
Pay frequencies	Weekly (52) · Bi-weekly (26) · Semi-monthly (24) · Monthly (12)
CPP1 employee rate	5.95%
CPP2 rate (YMPE→YAMPE)	4.00%
EI employee rate	1.63%
EI employer multiplier	1.4×

Generated from *lib/payroll/constants.ts* and *lib/payroll/engine.ts*. Numbers below 2026 finalization are projected from 2025 with light indexation; All values verified against CRA, CEIC and provincial 2026 sources.

1. Pay frequency math

Every per-period calculation starts by dividing the annual amount by **periods per year**. The reverse — annualizing — multiplies the period amount by the same factor so it can be compared against annual tax brackets and CRA exemptions.

Frequency	Periods / year	Formula (gross)
Weekly	52	$\text{annual_salary} / 52$
Bi-weekly	26	$\text{annual_salary} / 26$
Semi-monthly	24	$\text{annual_salary} / 24$
Monthly	12	$\text{annual_salary} / 12$

For hourly employees: $\text{gross} = \text{hourly_rate} \times \text{regular_hours} + \text{hourly_rate} \times 1.5 \times \text{overtime_hours} + \text{bonuses} + \text{vacation_accrual}$

2. Canada Pension Plan (CPP1)

CPP1 is the base tier. Pensionable earnings above the YBE and below YMPE are taxed at 5.95% (employee) matched 1:1 by the employer.

Constant	Value	Source
Year's Basic Exemption (YBE)	\$3,500.00	CRA — frozen
YMPE (CPP1 ceiling)	\$74,600.00	CRA 2026 confirmed
YAMPE (CPP2 ceiling)	\$85,000.00	CRA 2026 confirmed
Employee rate	5.95%	CRA
Annual max CPP1	\$4,230.45	$(\text{YMPE} - \text{YBE}) \times \text{rate}$
Annual max CPP2	\$416.00	$(\text{YAMPE} - \text{YMPE}) \times 4\%$

Per-period formula (lib/payroll/cpp.ts)

$\text{cpp} = \max(0, \text{pensionable} - \text{YBE}/\text{periods}) \times 5.95\%$ (capped so $\text{YTD} \leq \text{annual max}$)

Worked example — Ontario, salary \$92,000, bi-weekly

```
period_exempt = 3500 / 26 = 134.62
taxable_for_cpp = 3538.46 - 134.62 = 3403.84
contribution = 3403.84 × 0.0595 = 202.53
YTD remaining vs cap → if YTD CPP ≥ $4,230.45 → contribution = 0
```

3. CPP2 (second tier)

Earnings above YMPE up to YAMPE are taxed at 4% (employee + employer match). Once YTD pensionable earnings exceed YAMPE, CPP2 contribution = 0.

$\text{tier_overlap} = \max(0, \min(\text{year_end}, \text{YAMPE}) - \max(\text{year_start}, \text{YMPE}))$
 $\text{cpp2} = \text{tier_overlap} \times 4\%$ (capped at annual max)

```
Example – employee with YTD pensionable $70,000, this period adds $5,000:
year_start = 70,000 · year_end = 75,000 · YMPE = $74,600.00
overlap = min(75000, 85000) - max(70000, 74600) = 75000 - 74600 = 400
cpp2 = 400 × 4% = $16.00
```

4. Employment Insurance (EI)

EI premiums are deducted on insurable earnings until the annual maximum is hit. Employer pays 1.4× the employee premium.

Constant	Value
Maximum Insurable Earnings (MIE)	\$68,900.00
Employee rate	1.63%
Employer multiplier	1.4×
Annual max employee premium	\$1,123.07
Annual max employer premium	\$1,572.30

$ei_employee = insurable \times 1.63\%$ (capped at annual max)

$ei_employer = ei_employee \times 1.4$

Example – gross \$3,538.46 (Ontario, salary \$92k bi-weekly):

$ei_employee = 3,538.46 \times 0.0163 = \57.68

$ei_employer = \$57.68 \times 1.4 = \80.75

5. Federal income tax

Annual brackets for 2026 — **note the lowest rate is now 14%** (was 15% in 2025), per the Tax Cut for the Middle Class taking full effect. The engine annualizes the period gross, applies the brackets, subtracts the BPA credit, then divides back into the period.

Income up to	Marginal rate	Constant K
\$58,523.00	14.00%	\$0.00
\$117,045.00	20.50%	\$8,193.22
\$181,440.00	26.00%	\$20,190.23
\$258,482.00	29.00%	\$36,932.93
no ceiling	33.00%	\$59,275.11

Basic Personal Amount with phase-out:

income $\leq$ \$181,440.00 $\rightarrow$ BPA = \$16,452.00

\$181,440.00 < income < \$258,482.00 $\rightarrow$ linear from \$16,452.00 down to \$14,829.00

income $\geq$ \$258,482.00 $\rightarrow$ BPA = \$14,829.00

BPA credit = effective BPA $\times$ 14% — at \$92k income that's \$16,452.00 $\times$ 14% = \$2,303.28

Per-period formula (lib/payroll/federal-tax.ts)

```
annualized = period_gross  $\times$  periods
annual_tax = sum(bracket  $\times$  rate)
eff_BPA = federalBPA(annualized)  $\leftarrow$  phase-out aware
annual_net = max(0, annual_tax - eff_BPA  $\times$  14%)
period_fed = annual_net / periods
period_fed -= (CPP + CPP2 + EI)  $\times$  14%  $\leftarrow$  K2 (lowest rate)
```

Example – Ontario salary \$92,000 bi-weekly:

annualized = 3,538.46 $\times$ 26 = \$91,999.96

annual_tax = 58,523 $\times$ 14% + (92,000 – 58,523) $\times$ 20.5% = \$8,193.22 + \$6,862.78 = \$15,056.01

BPA credit = 16,452 $\times$ 14% = \$2,303.28

annual_net = \$15,056.01 – \$2,303.28 = \$12,752.73

period_fed = \$12,752.73 / 26 = \$490.49

(then reduced by ~\$36 of CPP+EI credit at 14% before final number)

6. Provincial income tax (9 provinces)

Same formula shape as federal — annualize, apply brackets, subtract provincial BPA credit, divide back. Ontario adds a surtax on top of provincial tax. Quebec is excluded.

Ontario (ON)

Income up to	Marginal rate
\$53,891.00	5.05%
\$107,785.00	9.15%
\$150,000.00	11.16%
\$220,000.00	12.16%
no ceiling	13.16%

BPA $\$12,989.00 \cdot \text{BPA credit} = \text{BPA} \times 5.05\% = \655.94

Surtax (applied to provincial tax above thresholds): 20.00% over \$5,818.00 · 36.00% over \$7,446.00

Alberta (AB)

Income up to	Marginal rate
\$61,200.00	8.00%
\$154,259.00	10.00%
\$185,111.00	12.00%
\$246,813.00	13.00%
\$370,220.00	14.00%
no ceiling	15.00%

BPA $\$22,769.00 \cdot \text{BPA credit} = \text{BPA} \times 8.00\% = \$1,821.52$

British Columbia (BC)

Income up to	Marginal rate
\$50,363.00	5.60%
\$100,728.00	7.70%
\$115,648.00	10.50%
\$140,430.00	12.29%
\$190,405.00	14.70%
\$265,545.00	16.80%
no ceiling	20.50%

BPA $\$13,216.00 \cdot \text{BPA credit} = \text{BPA} \times 5.60\% = \740.10

Manitoba (MB)

Income up to	Marginal rate
\$47,564.00	10.80%
\$101,200.00	12.75%
no ceiling	17.40%

BPA \$15,780.00 · BPA credit = $\text{BPA} \times 10.80\% = \$1,704.24$

Saskatchewan (SK)

Income up to	Marginal rate
\$54,532.00	10.50%
\$155,805.00	12.50%
no ceiling	14.50%

BPA \$20,381.00 · BPA credit = $\text{BPA} \times 10.50\% = \$2,140.01$

Nova Scotia (NS)

Income up to	Marginal rate
\$30,995.00	8.79%
\$61,991.00	14.95%
\$97,418.00	16.67%
\$157,124.00	17.50%
no ceiling	21.00%

BPA \$11,932.00 · BPA credit = $\text{BPA} \times 8.79\% = \$1,048.82$

New Brunswick (NB)

Income up to	Marginal rate
\$52,332.00	9.40%
\$104,666.00	14.00%
\$193,861.00	16.00%
no ceiling	19.50%

BPA \$13,664.00 · BPA credit = $\text{BPA} \times 9.40\% = \$1,284.42$

Prince Edward Island (PE)

Income up to	Marginal rate
\$32,656.00	9.80%
\$81,310.00	13.80%
no ceiling	16.70%

BPA \$14,525.00 · BPA credit = BPA × 9.80% = \$1,423.45

Newfoundland & Labrador (NL)

Income up to	Marginal rate
\$44,678.00	8.70%
\$89,355.00	14.50%
\$159,529.00	15.80%
\$223,340.00	17.80%
\$285,318.00	19.80%
\$570,635.00	20.80%
\$1,141,272.00	21.30%
no ceiling	21.80%

BPA \$15,000.00 · BPA credit = BPA × 8.70% = \$1,305.00

7. Vacation accrual & overtime

Vacation (lib/payroll/engine.ts)

Salaried employees: 0 (vacation paid as part of regular salary).

Hourly employees: $\text{vacation} = (\text{regular} + \text{overtime} + \text{bonus}) \times \text{vacation_percent}$. Default 4% (2 weeks).

Operator may set per-employee.

Overtime by province

When the operator records overtime hours, the engine pays $\text{hourly_rate} \times 1.5 \times \text{ot_hours}$. The thresholds below are advisory (UI hint) — the engine does not auto-compute OT from total hours yet.

Province	OT threshold (weekly hrs)	OT rate
Ontario	44	1.5× regular
Alberta	44	1.5× regular
British Columbia	40	1.5× regular
Manitoba	40	1.5× regular
Saskatchewan	40	1.5× regular
Nova Scotia	48	1.5× regular
New Brunswick	44	1.5× regular
Prince Edward Island	48	1.5× regular
Newfoundland & Labrador	40	1.5× regular

8. YTD enforcement (correctness layer)

Before every payroll run, **PayrollYTDService** folds all finalized + voided runs in the current tax year per employee and feeds the result into the engine. The CPP / CPP2 / EI services then cap the period contribution against the annual max:

$\text{contribution_capped} = \min(\text{contribution_uncapped}, \text{annual_max} - \text{YTD})$

Voided runs and their reversal entries are both included in the fold — they net to zero, so no special-case logic is needed in YTD. Preview/draft runs are excluded.

Caps for 2026: CPP1 = \$4,230.45 · CPP2 = \$416.00 · EI = \$1,123.07

9. Worked example A — \$80,000 salary, bi-weekly, all 9 provinces

Same gross. Different provincial tax + (in Ontario) surtax produce different net pay. These are the exact numbers NorthPay would put on a paystub.

Province	Gross / period	Fed	Prov	CPP	EI	Net / period	Net / yr
Ontario	\$3,076.92	\$364.34	\$171.33	\$175.07	\$50.15	\$2,316.03	\$60,216.78
Alberta	\$3,076.92	\$364.34	\$190.56	\$175.07	\$50.15	\$2,296.80	\$59,716.80
British Columbia	\$3,076.92	\$364.34	\$167.78	\$175.07	\$50.15	\$2,319.58	\$60,309.08
Manitoba	\$3,076.92	\$364.34	\$291.09	\$175.07	\$50.15	\$2,196.27	\$57,103.02
Saskatchewan	\$3,076.92	\$364.34	\$260.36	\$175.07	\$50.15	\$2,227.00	\$57,902.00
Nova Scotia	\$3,076.92	\$364.34	\$358.14	\$175.07	\$50.15	\$2,129.22	\$55,359.72
New Brunswick	\$3,076.92	\$364.34	\$288.78	\$175.07	\$50.15	\$2,198.58	\$57,163.08
Prince Edward Island	\$3,076.92	\$364.34	\$319.63	\$175.07	\$50.15	\$2,167.73	\$56,360.98
Newfoundland & Labrador	\$3,076.92	\$364.34	\$296.29	\$175.07	\$50.15	\$2,191.07	\$56,967.82

10. Worked example B — Ontario \$80,000 across 4 pay frequencies

Tiny differences in net per year come from rounding to cents each period and from how the per-period exemption (YBE/periods) divides differently. Annual totals agree within a few dollars.

Frequency	Periods/yr	Gross / period	Fed	Prov	CPP	EI	Net / period	Net / yr
weekly	52	\$1,538.46	\$182.17	\$85.66	\$87.53	\$25.08	\$1,158.02	\$60,217.04
biweekly	26	\$3,076.92	\$364.34	\$171.33	\$175.07	\$50.15	\$2,316.03	\$60,216.78
semi-monthly	24	\$3,333.33	\$394.70	\$185.60	\$189.66	\$54.33	\$2,509.04	\$60,216.96
monthly	12	\$6,666.67	\$789.41	\$371.21	\$379.31	\$108.67	\$5,018.07	\$60,216.84

11. Worked example C — hourly with overtime + vacation

Priya Sharma · British Columbia · \$38/hr · bi-weekly · 6% vacation · 4 OT hours this period

Line	Calculation	Amount
Regular pay	$38.0 \times 80 \text{ hrs}$	\$3,040.00
Overtime	$38.0 \times 1.5 \times 4 \text{ hrs}$	\$228.00
Earnings before vacation		\$3,268.00
Vacation accrual	$\$3,268.00 \times 6\%$	\$196.08
Gross pay		\$3,464.08
CPP1	$(\$3,464.08 - \text{YBE}/26) \times 5.95\%$	\$198.10
CPP2	earnings still below YMPE $\rightarrow 0$	\$0.00
EI	$\$3,464.08 \times 1.63\%$	\$56.46
Federal tax (after CPP/EI credit)		\$437.06
BC provincial tax		\$197.59
Total deductions		\$889.21
Net pay	gross – deductions	\$2,574.87

Employer also pays: CPP match \$198.10 + EI 1.4× = \$79.04. Total employer cost = gross + CPP + CPP2 + EI(1.4×) = \$3,741.22.

12. Paystub PDF — field mapping (lib/pdf/paystub.ts)

The downloadable paystub is rendered client-side with jsPDF (Letter, 612×792 pt). Each field on the PDF maps 1:1 to a line in the saved PayrollRun:

Field	Source
Employer name + address + BN + CRA account	CompanySettings (settings repo)
Pay period / pay date	PayrollRun.periodStart, .periodEnd, .payDate
Employee name, SIN, employment type, freq	PayrollLineResult.employee snapshot
Regular / Overtime / Bonus / Vacation	PayrollLineResult.regularPay, .overtimePay, .bonusAmount, .vacationAccrual
Gross pay	PayrollLineResult.grossPay
Federal / Provincial tax	PayrollLineResult.federalTax, .provincialTax
CPP (CPP1 + CPP2)	PayrollLineResult.cppEmployee + .cpp2Employee
EI	PayrollLineResult.eiEmployee
Net deposit	PayrollLineResult.netPay
YTD column (current vs YTD)	computeYTD() — folds all finalized + voided runs
Employer contributions footer	PayrollLineResult.cppEmployer, .cpp2Employer, .eiEmployer

13. T4 slip PDF — box mapping (lib/pdf/t4.ts)

Annual T4 slip is built per-employee by aggregating every finalized + voided run in the tax year. Box-to-source mapping:

Box	Field	Source
10	Province of employment	Employee.province
12	Social Insurance Number	Employee.sin
14	Employment income	Σ line.grossPay for the tax year
16	Employee CPP contributions	Σ line.cppEmployee
16A	Employee CPP2 contributions	Σ line.cpp2Employee
18	Employee EI premiums	Σ line.eiEmployee
22	Income tax deducted (fed + prov)	Σ (line.federalTax + line.provincialTax)
24	EI insurable earnings (capped at MIE)	min(Σ gross, \$68,900)
26	CPP pensionable earnings (capped at YMPE)	min(Σ gross, \$74,600)
28	Exempt — CPP / EI / PPIP	Empty checkboxes (not auto-detected)
50	RPP / DPSP registration number	Not modeled — blank
52	Pension adjustment	Not modeled — \$0.00
54	Business number	CompanySettings.businessNumber

Caps: Box 24 caps at MIE (\$68,900); Box 26 caps at YMPE (\$74,600). Above those amounts the box shows the cap, not the raw sum.

14. End-to-end run flow

```
PayrollView.handleRun()  
-> PayrollLifecycleService.finalize({ employees, period })  
* PayrollYTDSERVICE(runs).getYTDMap()  
* PayrollValidationService.validate(...) <- blocks on errors  
* runPayroll({ employees, ytdByEmployee, ... })  
- calculatePayrollLine per employee  
- calculateCPP(pensionable, periods, ytdCpp)  
- calculateCPP2(periodPens, ytdPens, ytdCpp2)  
- calculateEI(insurable, ytdEi)  
- calculateFederalTax(periodTaxable, periods)  
- calculateProvincialTax(province, periodTaxable, periods)  
- aggregate totals, compute inputHash  
* IPayrollRepository.save(run) <- rejects dup hash  
* AuditLogService.log('payroll.finalized')
```

Every step is pure (no side effects) until `save()` and `log()` at the very end. This keeps the math fully reproducible.

15. Glossary

Term	Meaning
YBE	Year's Basic Exemption (\$3,500). Earnings below this are CPP-exempt.
YMPE	Year's Maximum Pensionable Earnings. Top of CPP1 tier.
YAMPE	Year's Additional Maximum Pensionable Earnings. Top of CPP2 tier.
BPA	Basic Personal Amount. Tax-free baseline; converted to a credit at the lowest bracket rate.
MIE	Maximum Insurable Earnings for EI (\$68,900).
K2 credit	PDOC formula factor reducing federal tax by 15% of CPP+EI. Approximated here.
Surtax	Ontario only — extra provincial tax above two thresholds.
Pensionable	Earnings counted for CPP. In NorthPay = full gross.
Insurable	Earnings counted for EI. In NorthPay = full gross.